

Unit II — Lecture Notes

Rayleigh–Ritz Method • Galerkin’s Method
Applications of Fourier Transforms
M.Tech (ASE)

Contents

I	Direct Methods — The Rayleigh–Ritz Method	1
I.1	The Rayleigh–Ritz Method	2
I.2	Problem 1	2
I.3	Problem 2	4
I.4	Problem 3	7
II	Weighted Residual Method — Galerkin’s Method	9
II.1	The Method	9
II.2	Problem 1	10
II.3	Problem 2	12
II.4	Summary: The Galerkin Recipe	14
III	Applications of Fourier Transforms	15
III.1	Fourier Transform	15
III.2	Finite Fourier Transforms	16
III.3	Heat Conduction in a Metal Bar	16
III.4	Problem 1 – Ends kept at 0°C	17
III.5	Problem 2 – Insulated ends, $\ell = 6$	20
III.6	Problem 3 – Bar of length 5	23
III.7	Problem 4 – Insulated ends, $\ell = 4$	26
III.8	Summary	29

Part I: Direct Methods — The Rayleigh–Ritz Method

In this part, we will see the **boundary value problems** on the Euler’s differential equations of some variational problems given by the functional

$$I = \int_{x_1}^{x_2} f(x, y, y') dx. \quad (1)$$

I.1 The Rayleigh–Ritz Method

Rayleigh–Ritz method

In this method, we will construct a **trial solution** $\bar{y}(x)$ for the given boundary value problem using the **Lagrangian interpolation method**. Substitute this $\bar{y}(x)$ in (1) (i.e. in the functional), i.e.

$$I = \int_{x_1}^{x_2} f(x, \bar{y}, \bar{y}') dx,$$

Determining the constants

Then find out the required arbitrary constants $c_1, c_2, \dots$ with the help of the equations

$$\frac{\partial I}{\partial c_1} = 0, \quad \frac{\partial I}{\partial c_2} = 0, \quad \dots$$

and hence we get the solution of the problem.

I.2 Problem 1

Problem 1

Solve the boundary value problem $y'' - y + x = 0$ with $y(0) = 0$ and $y(1) = 0$, by extremizing the functional $I = \int_0^1 (2xy - y^2 - y'^2) dx$ by the Rayleigh–Ritz method.

Solution. Given the functional $I = \int_0^1 (2xy - y^2 - y'^2) dx$

$$\Rightarrow f = 2xy - y^2 - y'^2 \quad \Rightarrow \quad \frac{\partial f}{\partial y} = 2x - 2y, \quad \frac{\partial f}{\partial y'} = -2y'.$$

Verifying the BVP

$\therefore$ By Euler's equation, we have

$$\begin{aligned} \frac{\partial f}{\partial y} - \frac{d}{dx} \left(\frac{\partial f}{\partial y'} \right) &= 0 \\ \Rightarrow 2x - 2y - \frac{d}{dx} (-2y') &= 0 \\ \Rightarrow 2x - 2y + 2y'' &= 0 \end{aligned}$$

The BVP

$$\Rightarrow y'' + x - y = 0 \quad (2)$$

which is the required boundary value problem, with $y(0) = 0$ and $y(1) = 0$.

Trial Solution

Let us construct a trial solution for the problem (2) through the Lagrangian interpolation method. For

x	0 (x_0)	$\frac{1}{2}$ (x_1)	1 (x_2)
y	0 (y_0)	c (y_1)	0 (y_2)

∴ By Lagrange's interpolation formula,

$$\bar{y}(x) = \frac{(x - x_1)(x - x_2)}{(x_0 - x_1)(x_0 - x_2)} y_0 + \frac{(x - x_0)(x - x_2)}{(x_1 - x_0)(x_1 - x_2)} y_1 + \frac{(x - x_0)(x - x_1)}{(x_2 - x_0)(x_2 - x_1)} y_2$$

$$\begin{aligned} \Rightarrow \bar{y}(x) &= \frac{\left(x - \frac{1}{2}\right)(x - 1)}{\left(0 - \frac{1}{2}\right)(0 - 1)}(0) + \frac{(x - 0)(x - 1)}{\left(\frac{1}{2} - 0\right)\left(\frac{1}{2} - 1\right)}(c) \\ &\quad + \frac{(x - 0)\left(x - \frac{1}{2}\right)}{(1 - 0)\left(1 - \frac{1}{2}\right)}(0) \end{aligned}$$

$$\begin{aligned} \Rightarrow \bar{y}(x) &= 0 + \frac{x(x - 1)}{\frac{1}{2}\left(-\frac{1}{2}\right)}(c) + 0 \\ &= -4cx(x - 1) \\ &= 4cx(1 - x) \end{aligned}$$

$$\bar{y}(x) = c_1x(1 - x) = c_1(x - x^2)$$

$$\text{where } c_1 = 4c$$

Further $\bar{y}' = c_1(1 - 2x)$.

Substituting in I

Now substitute $\bar{y}$ and $\bar{y}'$ in the given functional:

$$\begin{aligned} I &= \int_0^1 (2x\bar{y} - \bar{y}^2 - \bar{y}'^2) dx \\ &= \int_0^1 \left[2x c_1(x - x^2) - c_1^2(x - x^2)^2 - c_1^2(1 - 2x)^2 \right] dx \\ &= \int_0^1 \left[2c_1(x^2 - x^3) - c_1^2(x^2 + x^4 - 2x^3) - c_1^2(1 + 4x^2 - 4x) \right] dx \\ &= 2c_1 \left[\frac{x^3}{3} - \frac{x^4}{4} \right]_0^1 - c_1^2 \left[\frac{x^3}{3} + \frac{x^5}{5} - \frac{2x^4}{4} \right]_0^1 - c_1^2 \left[x + \frac{4x^3}{3} - \frac{4x^2}{2} \right]_0^1 \end{aligned}$$

Minimizing over c_1

$$\begin{aligned}
&= 2c_1 \left(\left(\frac{1}{3} - \frac{1}{4} \right) - 0 \right) - c_1^2 \left(\left(\frac{1}{3} + \frac{1}{5} - \frac{1}{2} \right) - 0 \right) - c_1^2 \left(\left(1 + \frac{4}{3} - 2 \right) - 0 \right) \\
&= \frac{c_1}{6} - c_1^2 \left(\frac{10 + 6 - 15}{30} \right) - c_1^2 \left(\frac{1}{3} \right) = \frac{c_1}{6} - c_1^2 \left(\frac{1}{30} + \frac{10}{30} \right) = \frac{c_1}{6} - \frac{11 c_1^2}{30} \\
&\Rightarrow \quad I = \frac{c_1}{6} - \frac{11 c_1^2}{30}
\end{aligned}$$

To get c_1 , consider $\frac{\partial I}{\partial c_1} = 0$:

$$\frac{1}{6} - \frac{22c_1}{30} = 0 \quad \Rightarrow \quad \frac{22c_1}{30} = \frac{1}{6} \quad \Rightarrow \quad c_1 = \frac{5}{22}$$

Final Solution

$\therefore$ The solution is given by (the trial solution)

$$y(x) = c_1 x(1 - x) = \frac{5}{22} x(1 - x)$$

Required solution of the given boundary value problem

$$y(x) = \frac{5}{22} x(1 - x)$$

Correction to the handwritten notes

The handwritten notes had an arithmetic slip in evaluating the integrals ($\int_0^1 (x - x^2)^2 dx = \frac{1}{30}$ and $\int_0^1 (1 - 2x)^2 dx = \frac{1}{3}$), which led to $c_1 = -\frac{5}{38}$. With the correct values, $c_1 = \frac{5}{22}$ — the same answer obtained for this boundary value problem by Galerkin's method in Part II.

I.3 Problem 2**Problem 2**

Solve the boundary value problem $x^2 y^3 - x^3 y'^2 y'' = 0$ by the Rayleigh–Ritz method, extremizing the functional $\int_1^2 \frac{x^3}{y'^2} dx$ with $y(1) = 0$ and $y(2) = 3$.

Setting Up

Solution. Given the functional $I = \int_1^2 \frac{x^3}{y'^2} dx$

$$\Rightarrow \quad f = \frac{x^3}{y'^2} \quad \text{and } f \text{ is independent of } y.$$

$\therefore$ The respective Euler's equation is

$$\frac{d}{dx} \left(\frac{\partial f}{\partial y'} \right) = 0 \quad \Rightarrow \quad \frac{d}{dx} \left(\frac{y'^2 \cdot 0 - x^3 (2y')}{y'^4} \right) = 0$$

Deriving the BVP

$$\begin{aligned}
\Rightarrow \quad & \frac{d}{dx} \left(-\frac{2x^3 y'}{y'^4} \right) = 0 \quad \Rightarrow \quad \frac{d}{dx} \left(-\frac{2x^3}{y'^3} \right) = 0 \\
\Rightarrow \quad & \frac{(y'^3)(-6x^2) - (-2x^3)(3y'^2)y''}{(y'^3)^2} = 0 \\
\Rightarrow \quad & -6x^2 y'^3 + 6x^3 y'^2 y'' = 0
\end{aligned}$$

The BVP

$$\Rightarrow \quad -x^2 y'^3 + x^3 y'^2 y'' = 0 \quad (2)$$

which is the given boundary value problem.

Aside: $x^3 y'^2 y'' = x^2 y'^3 \Rightarrow xy'' = y' \Rightarrow x \frac{d^2 y}{dx^2} - \frac{dy}{dx} = 0 \Rightarrow (x^2 D^2 - xD)y = 0$.

Trial Solution

To get the trial solution $\bar{y}(x)$ for the above equation (2), we have

x	1 (x_0)	1.5 (x_1)	2 (x_2)
y	0 (y_0)	c (y_1)	3 (y_2)

$\therefore$ By Lagrange's interpolation formula,

$$\begin{aligned}
\bar{y}(x) &= \frac{(x-1.5)(x-2)}{(1-1.5)(1-2)}(0) + \frac{(x-1)(x-2)}{(1.5-1)(1.5-2)}(c) \\
&\quad + \frac{(x-1)(x-1.5)}{(2-1)(2-1.5)}(3) \\
&= 0 + \frac{x^2 - 3x + 2}{(0.5)(-0.5)}(c) + \frac{x^2 - 2.5x + 1.5}{1(0.5)}(3) \\
&= (x^2 - 3x + 2) \left(\frac{c}{-0.25} \right) + (x^2 - 2.5x + 1.5) \left(\frac{3}{0.5} \right) \\
&= (x^2 - 3x + 2)(-4c) + (x^2 - 2.5x + 1.5)(6)
\end{aligned}$$

$$\begin{aligned}
\bar{y}(x) &= x^2(-4c + 6) + x(12c - 15) + (-8c + 9) \\
\Rightarrow \quad \bar{y}'(x) &= 2x(-4c + 6) + (12c - 15) = x(12 - 8c) + (12c - 15) \\
\Rightarrow \quad \bar{y}'(x) &= c_1 x + c_2 \quad \text{where } c_1 = 12 - 8c \text{ and } c_2 = 12c - 15.
\end{aligned}$$

Substituting in I

Substituting in I :

$$I = \int_1^2 \frac{x^3}{\bar{y}'^2} dx = \int_1^2 \frac{x^3}{(c_1 x + c_2)^2} dx = \int_1^2 \frac{x^3}{c_1^2 x^2 + 2c_1 c_2 x + c_2^2} dx$$

Let $t = c_1 x + c_2 \Rightarrow x = \frac{t - c_2}{c_1}$, $dx = \frac{dt}{c_1}$; when $x = 1$, $t = c_1 + c_2$; when $x = 2$, $t = 2c_1 + c_2$.

$$\begin{aligned}
I &= \int_{c_1+c_2}^{2c_1+c_2} \frac{\left(\frac{t-c_2}{c_1}\right)^3}{t^2} \frac{dt}{c_1} = \int_{c_1+c_2}^{2c_1+c_2} \frac{1}{c_1^4} \frac{(t-c_2)^3}{t^2} dt \\
&= \frac{1}{c_1^4} \int_{c_1+c_2}^{2c_1+c_2} \left(\frac{t^3 - c_2^3 + 3tc_2^2 - 3t^2 c_2}{t^2} \right) dt
\end{aligned}$$

The Integral

$$\begin{aligned}
 &= \frac{1}{c_1^4} \int_{c_1+c_2}^{2c_1+c_2} \left(t - c_2^3 t^{-2} + \frac{3c_2^2}{t} - 3c_2 \right) dt \\
 &= \frac{1}{c_1^4} \left[\left(\frac{t^2}{2} \right)_{c_1+c_2}^{2c_1+c_2} - c_2^3 \left(\frac{t^{-1}}{-1} \right)_{c_1+c_2}^{2c_1+c_2} + 3c_2^2 \left(\log t \right)_{c_1+c_2}^{2c_1+c_2} - 3c_2 \left(t \right)_{c_1+c_2}^{2c_1+c_2} \right]
 \end{aligned}$$

Remark

= **lengthy calculations** (as noted in the class notes – the evaluation of $\partial I/\partial c_1 = 0$, $\partial I/\partial c_2 = 0$ becomes tedious).

Therefore (continuation of Problem 2)

Instead, the obtained differential equation (2) is solved *exactly* below.

Continuation: Exact Solution of the BVP

Obtained differential equation is

$$-x^2 y'^3 + x^3 y'^2 y'' = 0 \quad (2)$$

$$\begin{aligned}
 \text{i.e. } x^3 y'^2 y'' &= x^2 y'^3 & \Rightarrow & \quad x y'' = y' \\
 \text{i.e. } x y'' - y' &= 0 & \xrightarrow{\times \text{ by } x} & \quad x^2 y'' - x y' = 0
 \end{aligned}$$

$$\text{i.e. } (x^2 D^2 - x D) y = 0 \quad (3)$$

To convert (3) into a differential equation with **constant coefficients**, put

$$\begin{aligned}
 x &= e^z \text{ (or) } z = \log x; & \text{then} \\
 x D &= \theta, & x^2 D^2 = \theta(\theta - 1), & \text{where } \theta = \frac{d}{dz}.
 \end{aligned}$$

$$\begin{aligned}
 \therefore (3) \Rightarrow (\theta(\theta - 1) - \theta) y &= 0 \Rightarrow (\theta^2 - \theta - \theta) y = 0 \\
 &\Rightarrow (\theta^2 - 2\theta) y = 0
 \end{aligned} \quad (4)$$

which is a differential equation with constant coefficients.

Solution of (4): Auxiliary equation of (4) is

$$m^2 - 2m = 0 \Rightarrow m(m - 2) = 0 \Rightarrow m = 0, 2.$$

$\therefore$ Solution of (4) is

$$y = c_1 e^{0z} + c_2 e^{2z} \Rightarrow y = c_1 + c_2 e^{2z}.$$

$\therefore$ Solution of (3) is obtained by putting $x = e^z$ (or) $z = \log x$:

$$y = c_1 + c_2 x^2$$

Final Solution

i.e. $y(x) = c_1 + c_2x^2$ is the required solution of the given boundary value problem. (5)

Now apply $y(1) = 0$ in (5):

$$(5) \Rightarrow c_1 + c_2 = 0 \quad (6)$$

Now apply $y(2) = 3$ in (5):

$$(5) \Rightarrow 3 = c_1 + 4c_2 \quad (7)$$

Solving (6) & (7): $(7) - (6) \Rightarrow 3c_2 = 3$

$$c_2 = 1$$

$$c_1 = -1$$

Required solution

$$\therefore (5) \Rightarrow y(x) = -1 + x^2 \implies y(x) = x^2 - 1$$

I.4 Problem 3**Problem 3**

Solve the boundary value problem $y'' - 6x = 0$, by extremizing the functional $\int_0^1 (y'^2 + 12xy) dx$, given $y(0) = 0$ and $y(1) = 1$.

Solution. Given the functional

$$I = \int_0^1 (y'^2 + 12xy) dx \quad (1)$$

$$\Rightarrow f = y'^2 + 12xy \Rightarrow \frac{\partial f}{\partial y} = 12x, \quad \frac{\partial f}{\partial y'} = 2y'.$$

Euler's Equation

Then by Euler's equation, we have

$$\frac{\partial f}{\partial y} - \frac{d}{dx} \left(\frac{\partial f}{\partial y'} \right) = 0$$

$$\Rightarrow 12x - \frac{d}{dx} (2y') = 0$$

$$\Rightarrow 12x - 2y'' = 0$$

$$\Rightarrow 2y'' = 12x$$

The BVP

$$\Rightarrow y'' - 6x = 0 \quad (2)$$

which is the required boundary value problem, with $y(0) = 0$ and $y(1) = 1$.

Direct Integration

Equation (2) is a simple 2nd order ODE:

$$y'' = 6x$$

Integrating:

$$y' = \frac{6x^2}{2} + c_1 \quad \Rightarrow \quad y' = 3x^2 + c_1$$

Integrating again:

$$y = x^3 + c_1x + c_2 \quad \Rightarrow \quad y(x) = x^3 + c_1x + c_2 \quad (3)$$

which is its solution.

Now apply $y(0) = 0$ in (3) $\Rightarrow$ $0 = c_2$

Final Solution

$\therefore$ Solution (3) $\Rightarrow$

$$y(x) = x^3 + c_1x \quad (4)$$

is the updated solution.

Now apply $y(1) = 1$:

$$(4) \Rightarrow 1 = 1 + c_1 \quad \Rightarrow \quad c_1 = 0$$

Required solution

$$\therefore (4) \Rightarrow y(x) = x^3$$

Part II: Weighted Residual Method — Galerkin's Method

Variational Problem $\longrightarrow$ Boundary Value Problem

(Notes dated 01/12/2021)

II.1 The Method

Setting up the problem

From the **variational problem** we may have the **boundary value problem**

$$y'' = f[x, y, y'], \quad y(a) = A \text{ \& } y(b) = B.$$

Residual form

The above boundary value problem can be written in the *residual form*

$$R = \bar{y}'' - f[x, \bar{y}, \bar{y}']$$

Initial Solution

Trial (initial) solution

$$\bar{y}[x] = C_1 \phi_1[x] + C_2 \phi_2[x] + \dots$$

is the **initial solution**, where it must satisfy

$$\bar{y}[a] = A \quad \& \quad \bar{y}[b] = B.$$

The $\phi_i[x]$ are the **basis (weight) functions**, chosen so that every one of them vanishes at the boundaries.

Determining the Constants

Orthogonality conditions

To find $C_1, C_2, \dots$ we can use

$$\int_a^b R \phi_1[x] dx = 0, \quad \int_a^b R \phi_2[x] dx = 0, \quad \& \text{ so on.}$$

Substitute the values of $C_1, C_2, \dots$ back into $\bar{y}[x]$ to obtain the required approximate solution.

II.2 Problem 1

Problem 1

Extremize the functional

$$I = \int_0^1 [2xy - y^2 - y'^2] dx$$

with $y[0] = 0$ & $y[1] = 0$ using **Galerkin's Method**.

Solution — Step 1

$$f[x, y, y'] = 2xy - y^2 - y'^2 \quad (1)$$

Deriving the Differential Equation

By Euler's equation

$$\frac{\partial f}{\partial y} - \frac{d}{dx} \left[\frac{\partial f}{\partial y'} \right] = 0 \quad (2)$$

$$\begin{aligned} \frac{\partial f}{\partial y} &= 2x - 2y, & \frac{\partial f}{\partial y'} &= -2y' \\ [2x - 2y] - \frac{d}{dx} [-2y'] &= 0 & \implies & 2x - 2y + 2y'' = 0 \end{aligned}$$

Governing Differential Equation

Governing differential equation

$$\begin{aligned} x - y + y'' &= 0 & \implies & y'' - y + x = 0 \\ \boxed{y'' = y - x} & & \text{in the form } y'' &= f[x, y, y'] \end{aligned} \quad (3)$$

This is the boundary value problem to which Galerkin's method is applied.

The Residue

By Galerkin's Method

$$R = \bar{y}'' - f[x, \bar{y}, \bar{y}'] \implies \boxed{R = \bar{y}'' - \bar{y} + x}$$

Next: build the initial solution

The initial solution can be assumed with the help of **Lagrange's interpolation formula**, where $y[x]$ satisfies

$$y[0] = 0 \quad \& \quad y[1] = 0.$$

Lagrange's InterpolationTable of assumed values of $y[x]$

x	$x_0 = 0$	$x_1 = 0.5$	$x_2 = 1$
y	$y_0 = 0$	$y_1 = c$	$y_2 = 0$

Lagrange's interpolation formula

$$y[x] = \frac{(x - x_1)(x - x_2)}{(x_0 - x_1)(x_0 - x_2)} [y_0] + \frac{(x - x_0)(x - x_2)}{(x_1 - x_0)(x_1 - x_2)} [y_1] + \frac{(x - x_0)(x - x_1)}{(x_2 - x_0)(x_2 - x_1)} [y_2]$$

The Trial Function

$$y[x] = \frac{[x - 0.5][x - 1]}{(-0.5)(-1)} [0] + \frac{(x - 0)(x - 1)}{(0.5)(-0.5)} [c] + \frac{(x - 0)(x - 0.5)}{(1)(0.5)} [0]$$

$$y[x] = \frac{x(x - 1)}{\left(\frac{1}{2}\right)\left(-\frac{1}{2}\right)} c = -4c(x^2 - x)$$

Hence the initial solution is

$$\bar{y}[x] = 4c[x - x^2] \quad (4)$$

Forming the Residue

Derivatives of the trial function

$$\bar{y} = 4c[x - x^2], \quad \bar{y}' = 4c[1 - 2x], \quad \bar{y}'' = 4c[-2] = -8c$$

$$R = -8c - 4c[x - x^2] + x$$

Choose $\phi[x]$ from (4), i.e. from $\bar{y}[x]$, as $\phi[x] = x - x^2$ **Applying the Galerkin Condition**Apply $\int_0^1 R \phi[x] dx = 0$

$$\int_0^1 [-8c - 4c(x - x^2) + x][x - x^2] dx = 0$$

Expanding the integrand

$$\int_0^1 [-8cx + 8cx^2 - 4cx^2 + 4cx^3 + 4cx^3 - 4cx^4 + x^2 - x^3] dx = 0$$

Term-by-Term Integration

$$\begin{aligned}
 & -8c \left[\frac{x^2}{2} \right]_0^1 + 8c \left[\frac{x^3}{3} \right]_0^1 - 4c \left[\frac{x^3}{3} \right]_0^1 + 4c \left[\frac{x^4}{4} \right]_0^1 + 4c \left[\frac{x^4}{4} \right]_0^1 \\
 & - 4c \left[\frac{x^5}{5} \right]_0^1 + \left[\frac{x^3}{3} \right]_0^1 - \left[\frac{x^4}{4} \right]_0^1 = 0
 \end{aligned}$$

Evaluating the limits

$$-4c + \frac{8}{3}c - \frac{4}{3}c + 2c - \frac{4}{5}c + \frac{1}{3} - \frac{1}{4} = 0$$

Solving for c

$$\begin{aligned}
 c \left[-4 + \frac{8}{3} - \frac{4}{3} + 2 - \frac{4}{5} \right] &= -\frac{1}{3} + \frac{1}{4} = -\frac{1}{12} \\
 c \left[-2 + \frac{8}{3} - \frac{4}{3} - \frac{4}{5} \right] &= -\frac{1}{12} \\
 c \left[-2 + \frac{4}{3} - \frac{4}{5} \right] &= -\frac{1}{12} \\
 c \left[-\frac{22}{15} \right] &= -\frac{1}{12}
 \end{aligned}$$

$$c = \frac{1}{12} \times \frac{15}{22} = \frac{15}{264} \quad \Rightarrow \quad \boxed{c = \frac{5}{88}}$$

Final Answer

Substituting c in (4): $\bar{y}[x] = 4 \left[\frac{5}{88} \right] x [1 - x]$

Solution of Problem 1

$$\boxed{\bar{y}[x] = \frac{5}{22} x [1 - x]}$$

Check: $\bar{y}[0] = 0$ and $\bar{y}[1] = 0$ — both boundary conditions hold.

II.3 Problem 2**Problem 2**

Solve the boundary value problem

$$y'' + y + x = 0, \quad 0 \leq x \leq 1,$$

with $y(0) = 0$ & $y[1] = 0$ by **Galerkin's Method**. (1)

Initial SolutionAssumed values of $y[x]$ (similar to the previous sum)

x	0	0.5	1
y	0	c	0

By Lagrange's interpolation formula, $y[x] = -4c[x^2 - x]$.

Hence the initial solution is

$$\bar{y}[x] = 4c[x - x^2] \quad (2)$$

Forming the Residue

Derivatives of the trial function

$$\bar{y} = 4c[x - x^2], \quad \bar{y}' = 4c[1 - 2x], \quad \bar{y}'' = 4c[-2] = -8c$$

$$R = -8c + 4c[x - x^2] + x \quad (3)$$

And choose $\phi[x]$ from (2):

$$\phi[x] = [x - x^2]$$

Applying the Galerkin ConditionApply $\int_0^1 R \phi[x] dx = 0$

$$\int_0^1 [-8c + 4c(x - x^2) + x][x - x^2] dx = 0$$

Expanding the integrand

$$\int_0^1 [-8cx + 8cx^2 + 4cx^2 - 4cx^3 - 4cx^3 + 4cx^4 + x^2 - x^3] dx = 0$$

Term-by-Term Integration

$$\begin{aligned} & -8c \left[\frac{x^2}{2} \right]_0^1 + 8c \left[\frac{x^3}{3} \right]_0^1 + 4c \left[\frac{x^3}{3} \right]_0^1 - 4c \left[\frac{x^4}{4} \right]_0^1 - 4c \left[\frac{x^4}{4} \right]_0^1 \\ & + 4c \left[\frac{x^5}{5} \right]_0^1 + \left[\frac{x^3}{3} \right]_0^1 - \left[\frac{x^4}{4} \right]_0^1 = 0 \end{aligned}$$

Evaluating the limits

$$-4c + \left[\frac{8}{3}c + \frac{4}{3}c \right] - 2c + \frac{4}{5}c + \frac{1}{3} - \frac{1}{4} = 0$$

Solving for c

$$c \left[-6 + \frac{12}{3} + \frac{4}{5} \right] = -\frac{1}{12}$$

$$c \left[-6 + \frac{72}{15} \right] = -\frac{1}{12}$$

$$c \left[-\frac{18}{15} \right] = -\frac{1}{12}$$

$$c = \frac{1}{12} \times \frac{15}{18} \quad \Rightarrow \quad \boxed{c = \frac{5}{72}}$$

Final Answer

Substituting c in (2): $y[x] = 4 \left[\frac{5}{72} \right] [x - x^2]$

Solution of Problem 2

$$\boxed{y[x] = \frac{5}{18} [x - x^2]}$$

Check: $y[0] = 0$ and $y[1] = 0$ — both boundary conditions hold.

II.4 Summary: The Galerkin Recipe

1. Variational problem $\rightarrow$ B.V.P. (Euler's equation).
2. Residual $R = \bar{y}'' - f[x, \bar{y}, \bar{y}']$.
3. Trial $\bar{y}[x]$ meeting the boundary conditions.
4. Impose $\int_a^b R \phi_i[x] dx = 0$.
5. Solve for $C_1, C_2, \dots$; substitute back.

Results obtained

$$\text{P1: } \bar{y}[x] = \frac{5}{22} x(1-x) \quad \text{P2: } y[x] = \frac{5}{18} x(1-x)$$

Part III: Applications of Fourier Transforms

Finite Fourier Sine & Cosine Transforms and Heat Conduction in a Metal Bar

III.1 Fourier Transform

Fourier Transform (F–T)

Fourier Transform and its Inverse

If $f(x)$ is a continuous function defined in $(-\infty, \infty)$, then its **Fourier transform** is defined by

$$F(f(x)) = \frac{1}{\sqrt{2\pi}} \int_{-\infty}^{\infty} f(x) e^{isx} dx = F(s),$$

and its **inverse Fourier transform** is given by

$$f(x) = \frac{1}{\sqrt{2\pi}} \int_{-\infty}^{\infty} F(s) e^{-isx} ds.$$

Fourier Cosine and Sine Transforms

Observation

In the above definition, if $f(x)$ is an **even** function or an **odd** function, we get specifically a **cosine** or a **sine** function in the integral. Then the transforms are correspondingly called the **Fourier cosine transform (FCT)** or the **Fourier sine transform (FST)**.

Fourier Cosine Transform (FCT)

$$F_c(f(x)) = \sqrt{\frac{2}{\pi}} \int_0^{\infty} f(x) \cos sx dx = F_c(s),$$

and its inverse FCT is

$$f(x) = \sqrt{\frac{2}{\pi}} \int_0^{\infty} F_c(s) \cos sx ds.$$

Fourier Sine Transform (FST)

Fourier Sine Transform (FST)

$$F_s(f(x)) = \sqrt{\frac{2}{\pi}} \int_0^{\infty} f(x) \sin sx dx = F_s(s),$$

and its inverse FST is

$$f(x) = \sqrt{\frac{2}{\pi}} \int_0^{\infty} F_s(s) \sin sx ds.$$

Remark

The transforms above are defined on the *infinite* intervals $(-\infty, \infty)$ or $(0, \infty)$. Suppose instead that $f(x)$ is defined in $(0, \ell)$ — a **finite** interval. Then the respective transforms are called **finite Fourier cosine or sine transforms**.

III.2 Finite Fourier Transforms

Finite Fourier Cosine Transform (FFCT)

FFCT

$$F_c(f(x)) = \int_0^\ell f(x) \cos \frac{n\pi x}{\ell} dx = F_c(n)$$

Inverse FFCT

$$f(x) = \frac{1}{\ell} F_c(0) + \frac{2}{\ell} \sum_{n=1}^{\infty} F_c(n) \cos \frac{n\pi x}{\ell}$$

Finite Fourier Sine Transform (FFST)

FFST

$$F_s(f(x)) = \int_0^\ell f(x) \sin \frac{n\pi x}{\ell} dx = F_s(n)$$

Inverse FFST

$$f(x) = \frac{2}{\ell} \sum_{n=1}^{\infty} F_s(n) \sin \frac{n\pi x}{\ell}$$

III.3 Heat Conduction in a Metal Bar

Governing Partial Differential Equation

$$\frac{\partial u}{\partial t} = \alpha^2 \frac{\partial^2 u}{\partial x^2} \quad (1)$$

Meaning of the symbols

- $u(x, t)$ is the **heat (temperature) distribution** at a point x and at time t in the metal bar.
- α^2 is the **thermal diffusivity** of the substance.

Operational Results for FFCT and FFST

Idea

When we apply the FFCT (or the FFST) on equation (1), we get the following results. The bar occupies $x = 0$ to $x = \ell$.

Transform of the time derivative

$$(i) \quad F_c\left(\frac{\partial u}{\partial t}\right) = \frac{d}{dt} F_c(u), \quad (ii) \quad F_s\left(\frac{\partial u}{\partial t}\right) = \frac{d}{dt} F_s(u)$$

Operational Results (continued)**Transform of the second space derivative**

$$(iii) \quad F_c \left(\frac{\partial^2 u}{\partial x^2} \right) = (-1)^n \frac{\partial u}{\partial x}(\ell, t) - \frac{\partial u}{\partial x}(0, t) - \frac{n^2 \pi^2}{\ell^2} F_c(u)$$

$$(iv) \quad F_s \left(\frac{\partial^2 u}{\partial x^2} \right) = \frac{n\pi}{\ell} [u(0, t) - (-1)^n u(\ell, t)] - \frac{n^2 \pi^2}{\ell^2} F_s(u)$$

Which transform to use?

- Result (iv) needs the **values of** u at the two ends $\Rightarrow$ use **FFST** when $u(0, t)$ and $u(\ell, t)$ are given.
- Result (iii) needs the **values of** $\partial u / \partial x$ at the ends $\Rightarrow$ use **FFCT** when the ends are *insulated*, i.e. $u_x(0, t) = u_x(\ell, t) = 0$.

III.4 Problem 1 – Ends kept at 0°C**Problem 1**

A metal bar of length 4 cm, insulated at its both ends, has the temperature at its ends 0°C and the temperature inside the metal bar, initially, flows like $2x$ °C, $0 < x < 4$, and satisfies the heat equation

$$\frac{\partial u}{\partial t} = 2 \frac{\partial^2 u}{\partial x^2}.$$

Find the temperature at any point x and at a time t .

Given Data**Step 1: Write down the data**

Given $\ell = 4$ and

$$\frac{\partial u}{\partial t} = 2 \frac{\partial^2 u}{\partial x^2} \quad (1)$$

Also given

$$(i) \quad u(0, t) = 0 = u(4, t) \quad (\text{boundary conditions})$$

$$(ii) \quad u(x, 0) = 2x, \quad 0 < x < 4 \quad (\text{initial condition})$$

Choice of transform

The values of u itself are prescribed at the two ends, so we apply the **FFST**.

Apply FFST on (1)**Step 2: Transform the PDE**

Apply FFST on (1):

$$F_s \left(\frac{\partial u}{\partial t} \right) = 2 F_s \left(\frac{\partial^2 u}{\partial x^2} \right)$$

$$\Rightarrow \frac{d}{dt} F_s(u) = 2 \left\{ \frac{n\pi}{4} [u(0, t) - (-1)^n u(4, t)] - \frac{n^2 \pi^2}{4^2} F_s(u) \right\} \quad (\because \ell = 4)$$

Apply the boundary conditions**Step 2(a):** Put $F_s(u) = \bar{u}$ and use $u(0, t) = 0 = u(4, t)$

$$\Rightarrow \frac{d\bar{u}}{dt} = 2(0 - 0) - \frac{2n^2\pi^2}{16}\bar{u}$$

$$\Rightarrow \frac{d\bar{u}}{dt} = -\frac{n^2\pi^2}{8}\bar{u}$$

Observe

Both boundary terms vanish because the ends are held at 0°C , so only the term containing $F_s(u)$ survives.

A remark on the constant**The factor 2 must be carried through**

The factor $\alpha^2 = 2$ multiplies *both* terms inside the braces. Hence

$$2 \cdot \frac{n^2\pi^2}{4^2} = \frac{2n^2\pi^2}{16} = \frac{n^2\pi^2}{8},$$

so the denominator is **8** (not 16), and all the steps below use this value.

Working equation

$$\frac{d\bar{u}}{dt} = -\frac{n^2\pi^2}{8}\bar{u}$$

Solve the ODE**Step 3: Separate the variables and integrate**

$$\frac{d\bar{u}}{\bar{u}} = -\frac{n^2\pi^2}{8} dt \quad \Rightarrow \quad \int \frac{d\bar{u}}{\bar{u}} = -\frac{n^2\pi^2}{8} \int dt$$

$$\Rightarrow \log \bar{u} = -\frac{n^2\pi^2}{8} t + C \quad \Rightarrow \quad \bar{u} = e^{-\frac{n^2\pi^2}{8} t + C} = e^{-\frac{n^2\pi^2}{8} t} \cdot e^C$$

Transformed solution

$$\bar{u}(n, t) = A e^{-\frac{n^2\pi^2}{8} t} \quad (\because A \equiv e^C) \quad (2)$$

Notation

$\bar{u} = F_s(u)$, i.e. $\bar{u}(n, t) = F_s(u(x, t))$.

To find A **Step 4: Use the initial condition**

It is given that $u(x, 0) = 2x$, $0 < x < 4$. Apply the FFST on it:

$$F_s(u(x, 0)) = F_s(2x)$$

$$\bar{u}(n, 0) = \int_0^4 \underbrace{2x}_u \underbrace{\sin \frac{n\pi x}{4}}_v dx \quad (\because \ell = 4)$$

Using Bernoulli's rule $\int uv dx = uv_1 - u'v_2 + u''v_3 - \dots$:

$$= \left[2x \left(\frac{-\cos \frac{n\pi x}{4}}{\frac{n\pi}{4}} \right) - 2 \left(\frac{-\sin \frac{n\pi x}{4}}{\frac{n^2\pi^2}{16}} \right) + 0 \right]_0^4$$

Evaluating the integral**Step 5: Simplify**

$$\begin{aligned} &= -2 \cdot \frac{4}{n\pi} \left(x \cos \frac{n\pi x}{4} \right)_0^4 + 2 \cdot \frac{16}{n^2\pi^2} \left(\sin \frac{n\pi x}{4} \right)_0^4 \\ &= -\frac{8}{n\pi} (4 \cos n\pi - 0) + \frac{32}{n^2\pi^2} (\sin n\pi - 0) \\ &= -\frac{8}{n\pi} \cdot 4(-1)^n + \frac{32}{n^2\pi^2} (0 - 0) \quad (\because \sin n\pi = 0, \cos n\pi = (-1)^n) \end{aligned}$$

Transform of the initial temperature

$$\bar{u}(n, 0) = -\frac{32}{n\pi}(-1)^n \quad (3)$$

Determination of A **Step 6: Put $t = 0$ in (2)**

$$\text{Put } t = 0 \text{ in (2)} \implies \boxed{\bar{u}(n, 0) = A}$$

Therefore, by (3),

Value of A

$$A = -\frac{32}{n\pi}(-1)^n$$

The transformed solution**Step 7: Substitute A in (2)**

By (2), we have

$$\bar{u}(n, t) = A e^{-\frac{n^2 \pi^2}{8} t} \quad \implies \quad \bar{u}(n, t) = -\frac{32}{n\pi} (-1)^n e^{-\frac{n^2 \pi^2}{8} t}$$

Recovering u

$$\bar{u}(n, t) = F_s(u(x, t)) \quad \implies \quad u(x, t) = F_s^{-1}(\bar{u}(n, t))$$

where $f(x) = \frac{2}{\ell} \sum_{n=1}^{\infty} F_s(n) \sin \frac{n\pi x}{\ell}$.

Apply the inverse FFST**Step 8: Inverse transform**

$$\begin{aligned} u(x, t) &= \frac{2}{4} \sum_{n=1}^{\infty} \bar{u}(n, t) \sin \frac{n\pi x}{4} \\ &= \frac{1}{2} \sum_{n=1}^{\infty} \frac{(-1)^{n+1} 32}{n\pi} e^{-\frac{n^2 \pi^2}{8} t} \sin \frac{n\pi x}{4} \end{aligned}$$

Temperature distribution

$$u(x, t) = \frac{16}{\pi} \sum_{n=1}^{\infty} \frac{(-1)^{n+1}}{n} e^{-\frac{n^2 \pi^2}{8} t} \sin \frac{n\pi x}{4}$$

III.5 Problem 2 – Insulated ends, $\ell = 6$ **Problem 2**

A metal bar of length 6 cm has both of its ends **insulated**, so that

$$\frac{\partial u}{\partial x}(0, t) = 0 = \frac{\partial u}{\partial x}(6, t),$$

and the temperature inside the bar initially flows like $2x^\circ\text{C}$, $0 < x < 6$. If the temperature satisfies the heat equation $\frac{\partial u}{\partial t} = \frac{\partial^2 u}{\partial x^2}$, find the temperature distribution $u(x, t)$.

Note

In the manuscript only the data of this problem is recorded (the statement was not written out); the statement above is exactly the data used in the solution.

Given Data**Step 1: Write down the data**

Given $\ell = 6$. Also

$$\frac{\partial u}{\partial t} = \frac{\partial^2 u}{\partial x^2} \quad (1)$$

and

$$(i) \quad \frac{\partial u}{\partial x}(0, t) = 0 = \frac{\partial u}{\partial x}(6, t)$$

$$(ii) \quad u(x, 0) = 2x, \quad 0 < x < 6$$

Choice of transform

Here the *derivatives* of u are prescribed at the ends, so we apply the **FFCT**.

Apply FFCT on (1)**Step 2: Transform the PDE**

$$F_c\left(\frac{\partial u}{\partial t}\right) = F_c\left(\frac{\partial^2 u}{\partial x^2}\right)$$

$$\Rightarrow \frac{d}{dt}F_c(u) = (-1)^n \frac{\partial u}{\partial x}(6, t) - \frac{\partial u}{\partial x}(0, t) - \frac{n^2 \pi^2}{6^2} F_c(u)$$

Let $F_c(u) = \bar{u}$ and apply the boundary conditions:

$$\Rightarrow \frac{d\bar{u}}{dt} = 0 - 0 - \frac{n^2 \pi^2}{36} \bar{u} \quad \Rightarrow \quad \frac{d\bar{u}}{dt} = -\frac{n^2 \pi^2}{36} \bar{u}$$

Solve the ODE**Step 3: Separate the variables and integrate**

$$\begin{aligned} \frac{d\bar{u}}{\bar{u}} &= -\frac{n^2 \pi^2}{36} dt \quad \Rightarrow \quad \int \frac{d\bar{u}}{\bar{u}} = -\frac{n^2 \pi^2}{36} \int dt \\ \Rightarrow \log_e \bar{u} &= -\frac{n^2 \pi^2}{36} t + C \quad \Rightarrow \quad \bar{u} = e^{-\frac{n^2 \pi^2}{36} t} \cdot e^C \end{aligned}$$

Transformed solution

$$\bar{u}(n, t) = A e^{-\frac{n^2 \pi^2}{36} t} \quad (2)$$

To find A **Step 4: Use the initial condition**

Already we have $u(x, 0) = 2x$, $0 < x < 6$. Apply the FFCT on the above:

$$F_c(u(x, 0)) = \int_0^6 u(x, 0) \cos \frac{n\pi x}{6} dx = \int_0^6 \underbrace{2x}_u \underbrace{\cos \frac{n\pi x}{6}}_v dx$$

By Bernoulli's rule $= uv_1 - u'v_2 + u''v_3 - \dots$:

$$= \left[2x \left(\frac{\sin \frac{n\pi x}{6}}{\frac{n\pi}{6}} \right) - 2 \left(\frac{-\cos \frac{n\pi x}{6}}{\frac{n^2\pi^2}{36}} \right) + 0 \right]_0^6$$

Evaluating the integral**Step 5: Simplify**

$$= \frac{12}{n\pi} \left(x \sin \frac{n\pi x}{6} \right)_0^6 + \frac{72}{n^2\pi^2} \left(\cos \frac{n\pi x}{6} \right)_0^6$$

$$F_c(u(x, 0)) = \frac{12}{n\pi} (0 - 0) + \frac{72}{n^2\pi^2} ((-1)^n - 1)$$

Transform of the initial temperature

$$\bar{u}(n, 0) = \frac{72}{n^2\pi^2} ((-1)^n - 1) \quad (3)$$

Determination of A **Step 6: Put $t = 0$ in (2)**

$$\Rightarrow \bar{u}(n, 0) = A$$

Therefore, by (3),

Value of A

$$A = \frac{72}{n^2\pi^2} ((-1)^n - 1)$$

Transformed solution

$$\bar{u}(n, t) = \frac{72}{n^2\pi^2} ((-1)^n - 1) e^{-\frac{n^2\pi^2}{36}t} \quad (4)$$

Inverse FFCT**Step 7: Write the inversion formula**

By the inverse FFCT, we have

$$u(x, t) = \frac{1}{6} \bar{u}(0, 0) + \frac{2}{6} \sum_{n=1}^{\infty} \bar{u}(n, t) \cos \frac{n\pi x}{6} \quad (5)$$

Because

$$f(x) = \frac{1}{\ell} F_c(0) + \frac{2}{\ell} \sum_{n=1}^{\infty} F_c(n) \cos \frac{n\pi x}{\ell}$$

The $n = 0$ term must be computed separately — it is the mean value term.

Let us find $\bar{u}(0, 0)$

Step 8: The $n = 0$ term

$$\begin{aligned} \bar{u}(0, 0) &= [\bar{u}(n, 0)]_{n=0} = [F_c(u(x, 0))]_{n=0} = \left[\int_0^6 u(x, 0) \cos \frac{n\pi x}{6} dx \right]_{n=0} \\ &= \int_0^6 u(x, 0) dx = \int_0^6 2x dx = 2 \left(\frac{x^2}{2} \right)_0^6 = 36 \end{aligned}$$

Mean value term

$$\bar{u}(0, 0) = 36$$

Final answer

Step 9: Substitute in (5)

$$\begin{aligned} u(x, t) &= \frac{1}{6}(36) + \frac{1}{3} \sum_{n=1}^{\infty} \frac{72}{n^2 \pi^2} ((-1)^n - 1) e^{-\frac{n^2 \pi^2}{36} t} \cos \frac{n\pi x}{6} \\ &= 6 + 24 \sum_{n=1,3,5,\dots} \frac{-2}{n^2 \pi^2} e^{-\frac{n^2 \pi^2}{36} t} \cos \frac{n\pi x}{6} \end{aligned}$$

Temperature distribution

$$u(x, t) = 6 - 48 \sum_{n=1,3,5,\dots} \frac{1}{n^2 \pi^2} e^{-\frac{n^2 \pi^2}{36} t} \cos \frac{n\pi x}{6}$$

III.6 Problem 3 – Bar of length 5

Problem 3

A metal bar of length 5 cm having its ends' temperature at 0°C , and at $t = 0$ the temperature inside the metal bar kept at $5x^\circ\text{C}$, $0 < x < 5$. Find the temperature distribution of the metal bar at a point x and at a time t , if it satisfies the equation

$$\frac{\partial u}{\partial t} = 5 \frac{\partial^2 u}{\partial x^2}.$$

Note

In the manuscript this problem was left **unsolved** — only the data and the line “Now apply FFST on (1)” were written. The complete solution is worked out below.

Given Data**Step 1: Write down the data**

Given

$$\frac{\partial u}{\partial t} = 5 \frac{\partial^2 u}{\partial x^2} \quad \text{and} \quad \ell = 5. \quad (1)$$

Also given

$$(i) \quad u(0, t) = 0 = u(5, t)$$

$$(ii) \quad u(x, 0) = 5x, \quad 0 < x < 5$$

Choice of transformThe values of u are prescribed at both ends $\Rightarrow$ apply the **FFST**.**Apply FFST on (1)****Step 2: Transform the PDE**

$$\begin{aligned} F_s \left(\frac{\partial u}{\partial t} \right) &= 5 F_s \left(\frac{\partial^2 u}{\partial x^2} \right) \\ \Rightarrow \frac{d}{dt} F_s(u) &= 5 \left\{ \frac{n\pi}{5} [u(0, t) - (-1)^n u(5, t)] - \frac{n^2 \pi^2}{5^2} F_s(u) \right\} \\ & \quad (\because \ell = 5) \end{aligned}$$

Let $F_s(u) = \bar{u}$ and apply the boundary conditions $u(0, t) = 0 = u(5, t)$:

$$\begin{aligned} \Rightarrow \frac{d\bar{u}}{dt} &= 5(0 - 0) - \frac{5n^2\pi^2}{25} \bar{u} \\ \Rightarrow \frac{d\bar{u}}{dt} &= -\frac{n^2\pi^2}{5} \bar{u} \end{aligned}$$

Solve the ODE**Step 3: Separate the variables and integrate**

$$\begin{aligned} \frac{d\bar{u}}{\bar{u}} &= -\frac{n^2\pi^2}{5} dt \quad \Rightarrow \quad \int \frac{d\bar{u}}{\bar{u}} = -\frac{n^2\pi^2}{5} \int dt \\ \Rightarrow \log_e \bar{u} &= -\frac{n^2\pi^2}{5} t + C \quad \Rightarrow \quad \bar{u} = e^{-\frac{n^2\pi^2}{5} t} \cdot e^C \end{aligned}$$

Transformed solution

$$\bar{u}(n, t) = A e^{-\frac{n^2\pi^2}{5} t} \quad (\because A \equiv e^C) \quad (2)$$

To find A **Step 4: Use the initial condition**

It is given that $u(x, 0) = 5x$, $0 < x < 5$. Apply the FFST on it:

$$\bar{u}(n, 0) = \int_0^5 \underbrace{5x}_u \sin \underbrace{\frac{n\pi x}{5}}_v dx \quad (\because \ell = 5)$$

By Bernoulli's rule $= uv_1 - u'v_2 + u''v_3 - \dots$:

$$= \left[5x \left(\frac{-\cos \frac{n\pi x}{5}}{\frac{n\pi}{5}} \right) - 5 \left(\frac{-\sin \frac{n\pi x}{5}}{\frac{n^2\pi^2}{25}} \right) + 0 \right]_0^5$$

Evaluating the integral**Step 5: Simplify**

$$\begin{aligned} &= -5 \cdot \frac{5}{n\pi} \left(x \cos \frac{n\pi x}{5} \right)_0^5 + 5 \cdot \frac{25}{n^2\pi^2} \left(\sin \frac{n\pi x}{5} \right)_0^5 \\ &= -\frac{25}{n\pi} (5 \cos n\pi - 0) + \frac{125}{n^2\pi^2} (\sin n\pi - 0) \\ &= -\frac{125}{n\pi} (-1)^n + \frac{125}{n^2\pi^2} (0 - 0) \quad (\because \sin n\pi = 0, \cos n\pi = (-1)^n) \end{aligned}$$

Transform of the initial temperature

$$\bar{u}(n, 0) = -\frac{125}{n\pi} (-1)^n = \frac{125}{n\pi} (-1)^{n+1} \quad (3)$$

Determination of A **Step 6: Put $t = 0$ in (2)**

$$\Rightarrow \bar{u}(n, 0) = A$$

Therefore, by (3),

Value of A

$$A = \frac{125}{n\pi} (-1)^{n+1}$$

Transformed solution

$$\bar{u}(n, t) = \frac{125}{n\pi} (-1)^{n+1} e^{-\frac{n^2\pi^2}{5}t} \quad (4)$$

Apply the inverse FFST**Step 7: Inverse transform**

$$\begin{aligned}
 u(x, t) &= \frac{2}{5} \sum_{n=1}^{\infty} \bar{u}(n, t) \sin \frac{n\pi x}{5} \\
 &= \frac{2}{5} \sum_{n=1}^{\infty} \frac{125}{n\pi} (-1)^{n+1} e^{-\frac{n^2\pi^2}{5}t} \sin \frac{n\pi x}{5}
 \end{aligned}$$

Temperature distribution

$$u(x, t) = \frac{50}{\pi} \sum_{n=1}^{\infty} \frac{(-1)^{n+1}}{n} e^{-\frac{n^2\pi^2}{5}t} \sin \frac{n\pi x}{5}$$

Check**Verification**

- At $t = 0$: $u(x, 0) = \frac{50}{\pi} \sum_{n \geq 1} \frac{(-1)^{n+1}}{n} \sin \frac{n\pi x}{5} = 5x$ for $0 < x < 5$, which is the half-range sine (Fourier) series of $5x$. ✓
- At $x = 0$ and $x = 5$ every $\sin \frac{n\pi x}{5}$ vanishes, so $u(0, t) = 0 = u(5, t)$. ✓
- Each term satisfies $u_t = 5u_{xx}$ since $5 \left(\frac{n\pi}{5}\right)^2 = \frac{n^2\pi^2}{5}$. ✓

III.7 Problem 4 – Insulated ends, $\ell = 4$ **Problem 4**

Solve

$$\frac{\partial u}{\partial t} = \frac{\partial^2 u}{\partial x^2},$$

given that

- (i) $\frac{\partial u}{\partial x}(0, t) = 0 = \frac{\partial u}{\partial x}(4, t)$
- (ii) $u(x, 0) = x(4 - x), \quad 0 < x < 4$

Apply FFCT on (1)**Step 1: Data and transformation**Given $\ell = 4$ and

$$\frac{\partial u}{\partial t} = \frac{\partial^2 u}{\partial x^2} \tag{1}$$

Apply FFCT on (1):

$$\begin{aligned}
 F_c\left(\frac{\partial u}{\partial t}\right) &= F_c\left(\frac{\partial^2 u}{\partial x^2}\right) \\
 \Rightarrow \frac{d}{dt}F_c(u) &= (-1)^n \frac{\partial u}{\partial x}(4, t) - \frac{\partial u}{\partial x}(0, t) - \frac{n^2\pi^2}{4^2}F_c(u)
 \end{aligned}$$

Apply the boundary conditions**Step 2:** Use $u_x(0, t) = 0 = u_x(4, t)$ and put $F_c(u) = \bar{u}$

$$\begin{aligned} \frac{d\bar{u}}{dt} &= 0 - 0 - \frac{n^2\pi^2}{16} \bar{u} \quad \Rightarrow \quad \frac{d\bar{u}}{dt} = -\frac{n^2\pi^2}{16} \bar{u} \\ \Rightarrow \frac{d\bar{u}}{\bar{u}} &= -\frac{n^2\pi^2}{16} dt \quad \Rightarrow \quad \log_e \bar{u} = -\frac{n^2\pi^2}{16} t + C \\ \Rightarrow \bar{u} &= e^{-\frac{n^2\pi^2}{16} t} \cdot e^C \end{aligned}$$

Transformed solution

$$\bar{u}(n, t) = A e^{-\frac{n^2\pi^2}{16} t} \quad (2)$$

To find A**Step 3:** Use the initial conditionWe have $u(x, 0) = x(4 - x) = 4x - x^2$, $0 < x < 4$. Apply the FFCT on the above:

$$F_c(u(x, 0)) = \int_0^4 u(x, 0) \cos \frac{n\pi x}{4} dx = \int_0^4 \underbrace{(4x - x^2)}_u \underbrace{\cos \frac{n\pi x}{4}}_v dx$$

Here $u = 4x - x^2$ is differentiated repeatedly ($u' = 4 - 2x$, $u'' = -2$, $u''' = 0$) and $v = \cos \frac{n\pi x}{4}$ is integrated repeatedly.**Bernoulli's expansion****Step 3(a):** Apply $\int uv dx = uv_1 - u'v_2 + u''v_3 - \dots$

$$\begin{aligned} &= \left[(4x - x^2) \left(\frac{\sin \frac{n\pi x}{4}}{\frac{n\pi}{4}} \right) - (4 - 2x) \left(\frac{-\cos \frac{n\pi x}{4}}{\frac{n^2\pi^2}{16}} \right) \right. \\ &\quad \left. + (-2) \left(\frac{-\sin \frac{n\pi x}{4}}{\frac{n^3\pi^3}{64}} \right) \right]_0^4 \end{aligned}$$

Evaluating the integral**Step 4: Simplify**

$$\begin{aligned}
&= \frac{4}{n\pi} \left[(4x - x^2) \sin \frac{n\pi x}{4} \right]_0^4 + \frac{16}{n^2\pi^2} \left[(4 - 2x) \cos \frac{n\pi x}{4} \right]_0^4 \\
&\quad + \frac{128}{n^3\pi^3} \left[\sin \frac{n\pi x}{4} \right]_0^4 \\
F_c(u(x, 0)) &= \frac{4}{n\pi} (0 - 0) + \frac{16}{n^2\pi^2} (-4(-1)^n - 4(1)) + \frac{128}{n^3\pi^3} (0 - 0)
\end{aligned}$$

Transform of the initial temperature

$$\bar{u}(n, 0) = -\frac{64}{n^2\pi^2} ((-1)^n + 1)$$

Determination of A**Step 5: Put $t = 0$ in (2)**

$$\Rightarrow \bar{u}(n, 0) = A$$

Value of A

$$A = -\frac{64}{n^2\pi^2} ((-1)^n + 1)$$

Transformed solution – by (2)

$$\bar{u}(n, t) = -\frac{64}{n^2\pi^2} (1 + (-1)^n) e^{-\frac{n^2\pi^2}{16}t} \quad (3)$$

Let us find $\bar{u}(0, 0)$ **Step 6: The $n = 0$ term**

$$\begin{aligned}
\bar{u}(0, 0) &= [\bar{u}(n, 0)]_{n=0} = [F_c(u(x, 0))]_{n=0} = \left[\int_0^4 u(x, 0) \cos \frac{n\pi x}{4} dx \right]_{n=0} \\
&= \int_0^4 (4x - x^2) dx \quad (\because n = 0) \\
&= 4 \left(\frac{x^2}{2} \right)_0^4 - \left(\frac{x^3}{3} \right)_0^4 = 2(16 - 0) - \left(\frac{64}{3} - 0 \right) = 32 - \frac{64}{3} = \frac{96 - 64}{3}
\end{aligned}$$

Mean value term

$$\bar{u}(0, 0) = \frac{32}{3} \quad (4)$$

Inverse FFCT

Step 7: Inversion

By the inverse FFCT, we have

$$\begin{aligned} u(x, t) &= \frac{1}{4} \bar{u}(0, 0) + \frac{2}{4} \sum_{n=1}^{\infty} \bar{u}(n, t) \cos \frac{n\pi x}{4} \\ &= \frac{1}{4} \left(\frac{32}{3} \right) + \frac{1}{2} \sum_{n=1}^{\infty} \frac{-64}{n^2 \pi^2} (1 + (-1)^n) e^{-\frac{n^2 \pi^2}{16} t} \cos \frac{n\pi x}{4} \end{aligned}$$

Odd n vanish

For odd n , $1 + (-1)^n = 0$; for even n , $1 + (-1)^n = 2$. Hence only $n = 2, 4, 6, \dots$ survive.

Final answer

Temperature distribution

$$u(x, t) = \frac{8}{3} - 64 \sum_{n=2,4,6,\dots} \frac{1}{n^2 \pi^2} e^{-\frac{n^2 \pi^2}{16} t} \cos \frac{n\pi x}{4}$$

Interpretation

As $t \rightarrow \infty$ every exponential dies out and $u \rightarrow \frac{8}{3}$ — the average of the initial temperature $x(4-x)$ over $0 < x < 4$. This is exactly what is expected for a bar with **both ends insulated**: the total heat is conserved and finally distributes itself uniformly.

III.8 Summary

Summary of the Method

Working rule

1. Identify ℓ and the type of boundary conditions.
2. If u is prescribed at the ends use **FFST**; if u_x is prescribed (insulated ends) use **FFCT**.
3. Transform the PDE $\Rightarrow$ first order ODE in t for $\bar{u}(n, t)$.
4. Solve the ODE: $\bar{u}(n, t) = A e^{-\alpha^2 n^2 \pi^2 t / \ell^2}$.
5. Put $t = 0$ and use the transform of the initial condition to get A .
6. Apply the inverse transform to recover $u(x, t)$ (remember the extra $\frac{1}{\ell} F_c(0)$ term for the FFCT).

Summary of the Answers

All four problems

$$\mathbf{P1:} \quad u(x, t) = \frac{16}{\pi} \sum_{n=1}^{\infty} \frac{(-1)^{n+1}}{n} e^{-\frac{n^2 \pi^2}{8} t} \sin \frac{n \pi x}{4}$$

$$\mathbf{P2:} \quad u(x, t) = 6 - 48 \sum_{n=1,3,5,\dots} \frac{1}{n^2 \pi^2} e^{-\frac{n^2 \pi^2}{36} t} \cos \frac{n \pi x}{6}$$

$$\mathbf{P3:} \quad u(x, t) = \frac{50}{\pi} \sum_{n=1}^{\infty} \frac{(-1)^{n+1}}{n} e^{-\frac{n^2 \pi^2}{5} t} \sin \frac{n \pi x}{5}$$

$$\mathbf{P4:} \quad u(x, t) = \frac{8}{3} - 64 \sum_{n=2,4,6,\dots} \frac{1}{n^2 \pi^2} e^{-\frac{n^2 \pi^2}{16} t} \cos \frac{n \pi x}{4}$$